

PAST EXAMINATION QUESTIONS - 2018

(2018 Stage 2 Examinations, question 3)

(a) [12 Marks]

If $z = xe^y + \sin(xy)$ and $x + \ln|x| = \sin(y)$, then state the general expression giving $\frac{dz}{dx}$ in terms of $\frac{dy}{dx}$ and the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and use it to evaluate $\frac{dz}{dx}$ for this particular example.

(b) [13 Marks]

Reverse the order of integration in order to evaluate the integral $\int_0^2 \int_{y^2}^4 y \sin(x^2) dx dy$.

Sketch plots illustrating the shape of the region of integration and the orientation of the infinitesimal area elements for both the original and the reversed order of integration.

Solution

(a) If $z = xe^y + \sin(xy)$ and $x + \ln|x| = \sin(y)$ then z is indirectly a function of x alone.

In general, if $z = f(x, y)$ and $x = g(t)$ and $y = h(t)$, the chain rules states that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

But in the case where x and y are related implicitly we can write instead that the total differential is given by the expression

$$\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \frac{dy}{dx}.$$

In this particular case

$$\frac{\partial z}{\partial x} = e^y + y \cos(xy)$$

$$\frac{\partial z}{\partial y} = xe^y + x \cos(xy)$$

and implicit differentiation of $x + \ln|x| = \sin(y)$ gives

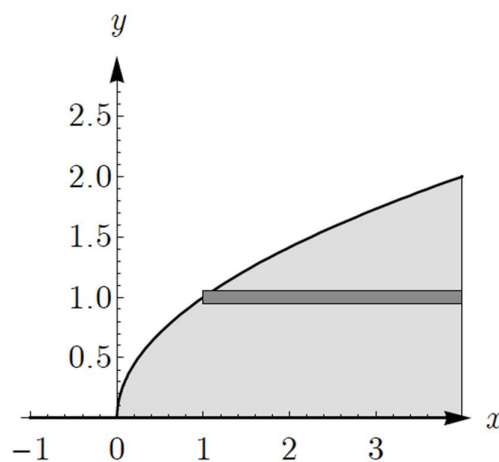
$$1 + \frac{1}{x} = \frac{dy}{dx} \cos(y) \Rightarrow \frac{dy}{dx} = \frac{1+x}{x \cos(y)}$$

Hence,

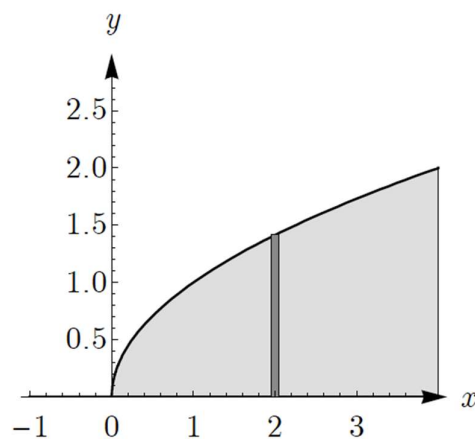
$$\begin{aligned}\frac{dz}{dx} &= \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \frac{dy}{dx} = e^y + y \cos(xy) + (xe^y + x \cos(xy)) \frac{1+x}{x \cos(y)} \\ &= e^y + y \cos(xy) + (e^y + \cos(xy)) \frac{(1+x)}{\cos(y)}\end{aligned}$$

(b) Let $I = \int_0^2 \int_{y^2}^4 y \sin(x^2) \, dx \, dy$.

This is a double integral representing an integration over a region whose x -limits are defined by the curve $x = y^2$, the x -axis and the vertical line $x = 4$ - the x -integration is carried out first, corresponding to horizontal infinitesimal area elements:



Reversing the order of integration will produce the integral $I = \int_0^4 \int_0^{\sqrt{x}} y \sin(x^2) \, dy \, dx$ - here the y -integration is carried out first, corresponding to vertical infinitesimal area elements, and the limits are defined by the line $y = 0$ (the x -axis) and the curve $y = \sqrt{x}$:



The integral can be evaluated as

$$\begin{aligned}
 I &= \int_0^4 \int_0^{\sqrt{x}} y \sin(x^2) dy dx = \int_0^4 \sin(x^2) \left(\int_0^{\sqrt{x}} y dy \right) dx \\
 &= \int_0^4 \sin(x^2) \left[\frac{y^2}{2} \right]_0^{\sqrt{x}} dx \\
 &= \frac{1}{2} \int_0^4 x \sin(x^2) dx \\
 &= \frac{1}{4} \left[-\cos(x^2) \right]_0^4 \\
 &= \frac{1}{4} [-\cos(16) + 1] = \frac{1}{4} [1 - \cos(16)]
 \end{aligned}$$

(2018 Stage 2 Examinations, question 4)

(a) [12 Marks]

Using 2D polar co-ordinates, obtain the area enclosed by the curve

$$r = a(\cos(\theta) + 1) \quad \text{for } 0 \leq \theta < 2\pi.$$

(b) [13 Marks]

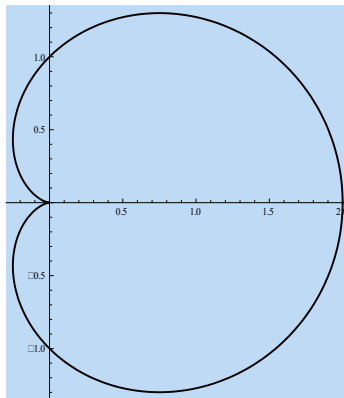
Evaluate the integral

$$\iiint_V z^2 dx dy dz$$

over the volume V contained within the sphere of radius a centred at the origin.

Solution

(a) The area within the cardioid curve $r = a(1 + \cos(\theta))$:



is scanned out as θ varies completely over the range $[0, 2\pi]$ and as r varies from zero up to the variable upper limit $r = a(1 + \cos(\theta))$.

For a simple area, the function to be integrated is $f(x, y) = 1$ and, under the transformation to polar coordinates, this is still a constant $f(r, \theta) = 1$.

The polar area element is $r \, dr \, d\theta$ and hence, in polar terms, the required integral is

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^{a(1+\cos(\theta))} 1 \, r \, dr \, d\theta = \int_0^{2\pi} \left(\int_0^{a(1+\cos(\theta))} r \, dr \right) d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^{a(1+\cos(\theta))} d\theta \\ &= \frac{a^2}{2} \int_0^{2\pi} (1 + \cos(\theta))^2 d\theta \\ &= \frac{a^2}{2} \int_0^{2\pi} (1 + 2\cos(\theta) + \cos^2(\theta)) d\theta \\ &= \frac{a^2}{2} \int_0^{2\pi} \left(1 + 2\cos(\theta) + \frac{1}{2}(1 + \cos(2\theta)) \right) d\theta \\ &= \frac{a^2}{4} \int_0^{2\pi} (3 + 4\cos(\theta) + \cos(2\theta)) d\theta \end{aligned}$$

Since $\cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta))$.

Hence

$$I = \frac{a^2}{4} \left[3\theta + 4\sin(\theta) + \frac{1}{2}\sin(2\theta) \right]_0^{2\pi} = \frac{a^2}{4} 6\pi = \frac{3\pi a^2}{2}$$

(b) To evaluate the integral $\iiint_V z^2 \, dx \, dy \, dz$ over a sphere of radius a centred at the origin, transform to spherical polar coordinates using

$$\begin{aligned} x &= r \sin(\theta) \cos(\phi) \\ y &= r \sin(\theta) \sin(\phi) \\ z &= r \cos(\theta) \end{aligned}$$

We have

- the volume element: $dx \, dy \, dz = r^2 \sin(\theta) \, dr \, d\theta \, d\phi$
- the integrand: $f(r, \theta, \phi) = r^2 \cos^2(\theta)$
- and the limits defined by the ranges $r \in [0, a]$, $\theta \in [0, \pi]$, and $\phi \in [0, 2\pi]$

Hence

$$\begin{aligned}
 I &= \iiint_V z^2 dz dy dx = \int_0^{2\pi} \int_0^\pi \int_0^a \left(r^2 \cos^2(\theta) \right) r^2 \sin(\theta) dr d\theta d\phi \\
 &= \int_0^{2\pi} \int_0^\pi \int_0^a r^4 \sin(\theta) \cos^2(\theta) dr d\theta d\phi \\
 &= \int_0^{2\pi} \int_0^\pi \sin(\theta) \cos^2(\theta) \int_0^a r^4 dr d\theta d\phi
 \end{aligned}$$

The integrand is a simple product of separate functions with constant limits - the integral can be broken down into three independent parts, giving

$$\begin{aligned}
 I &= \int_0^{2\pi} \int_0^\pi \sin(\theta) \cos^2(\theta) \int_0^a r^4 dr d\theta d\phi \\
 &= \left(\int_0^{2\pi} d\phi \right) \left(\int_0^\pi \sin(\theta) \cos^2(\theta) d\theta \right) \left(\int_0^a r^4 dr \right) \\
 &= \left[\phi \right]_0^{2\pi} \left[\left(-\frac{1}{3} \cos^3(\theta) \right) \right]_0^\pi \left[\frac{1}{5} r^5 \right]_0^a \\
 &= (2\pi) \left(\frac{2}{3} \right) \left(\frac{a^5}{5} \right) = \frac{4\pi a^5}{15}
 \end{aligned}$$

PAST EXAMINATION QUESTIONS - 2019

(2019 Stage 2 Examinations, question 3)

(a) [13 Marks]

For the surface represented by

$$z = xy^2 - 3xy + 3x^2 + 2x$$

obtain the coordinates of the stationary points and determine their nature.

(b) [12 Marks]

State the general expression for a Taylor series expansion of $f(x, y)$ about the point (a, b) up to and including second-order terms.

If $f(x, y) = xe^{(x^2y)}$, obtain a Taylor series expansion of $f(x, y)$ about the point $(1, 1)$ up to and including second-order terms.

Solution

(a) If $z = xy^2 - 3xy + 3x^2 + 2x$ then

$$\frac{\partial z}{\partial x} = y^2 - 3y + 6x + 2, \text{ so that } \frac{\partial z}{\partial x} = 0 \Rightarrow y^2 - 3y + 6x + 2 = 0 \text{ (condition 1)}$$

$$\text{and } \frac{\partial z}{\partial y} = 2xy - 3x, \text{ so that } \frac{\partial z}{\partial y} = 0 \Rightarrow x(2y - 3) = 0 \Rightarrow x = 0 \text{ or } y = \frac{3}{2} \text{ (condition 2).}$$

Setting $x = 0$ in condition 1 gives

$$3y^2 - 6y - 9 = 0 \Rightarrow y^2 - 3y + 2 = 0 \Rightarrow (y - 1)(y - 2) = 0 \Rightarrow y = 1 \text{ or } y = 2$$

$$\text{Likewise, setting } y = \frac{3}{2} \text{ in condition 1 gives } \frac{9}{4} - \frac{9}{2} + 6x + 2 = 0 \Rightarrow x = \frac{1}{24}$$

Hence there are three stationary points at:

$$(0, 1), (0, 2) \text{ and } \left(\frac{1}{24}, \frac{3}{2}\right).$$

In general for this function

$$\frac{\partial^2 z}{\partial x^2} = 6,$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2y - 3,$$

$$\text{and } \frac{\partial^2 z}{\partial y^2} = 2x.$$

At $(0,1)$ the value of the discriminant is $\Delta(0,1) = \begin{vmatrix} 6 & -1 \\ -1 & 0 \end{vmatrix} = -1$.

Since $\Delta < 0$ the stationary point at $(0,1)$ is a saddle point.

At $(0,2)$ the value of the discriminant is $\Delta(0,2) = \begin{vmatrix} 6 & 1 \\ 1 & 0 \end{vmatrix} = -1$.

Since $\Delta < 0$ the stationary point at $(0,2)$ is a saddle point.

At $\left(\frac{1}{24}, \frac{3}{2}\right)$ the value of the discriminant is $\Delta\left(\frac{1}{24}, \frac{3}{2}\right) = \begin{vmatrix} 6 & 0 \\ 0 & 1/12 \end{vmatrix} = \frac{1}{2}$.

Since $\Delta > 0$ and $\frac{\partial^2 f}{\partial x^2} > 0$, the stationary point at $\left(\frac{1}{24}, \frac{3}{2}\right)$ is a local minimum.

(b) The general Taylor series expansion of $f(x, y)$ about the point (a, b) up to and including second-order terms is

$$\begin{aligned} f(x, y) &= f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b) \\ &\quad + \frac{1}{2!}(x - a)^2 f_{xx}(a, b) + (x - a)(y - b)f_{xy}(a, b) + \frac{1}{2!}(y - b)^2 f_{yy}(a, b) \end{aligned}$$

If $f(x, y) = xe^{x^2y}$, then $f(1,1) = e$

$$f_x = 2x^2ye^{x^2y} + e^{x^2y}, \quad \text{so that } f_x(1,1) = 3e$$

$$f_y = x^3e^{x^2y}, \quad \text{so that } f_y(1,1) = e$$

$$f_{xx} = 4x^3y^2e^{x^2y} + 4xye^{x^2y} + 2xye^{x^2y}, \quad \text{so that } f_{xx}(1,1) = 10e$$

$$f_{yy} = x^5e^{x^2y}, \quad \text{so that } f_{yy}(1,1) = e$$

$$f_{xy} = 3x^2e^{x^2y} + 2x^4ye^{x^2y}, \quad \text{so that } f_{xy}(1,1) = 5e$$

Hence

$$f(x, y) \approx e + 3e(x - 1) + e(y - 1) + 5e(x - 1)^2 + 5e(x - 1)(y - 1) + 0.5e(y - 1)^2$$

(2019 Stage 2 Examinations, question 4)

(a) [12 Marks]

By reversing the order of integration, evaluate the integral

$$\int_0^1 \int_0^{\arccos(y)} \tan(x) dx dy$$

over the region defined by the lines $y = 0$ and $y = 1$, and $x = 0$ and $x = \arccos(y)$.

Sketch plots illustrating the shape of the region of integration and the orientation of the infinitesimal area elements for both the original and the reversed order of integration.

(b) [13 Marks]

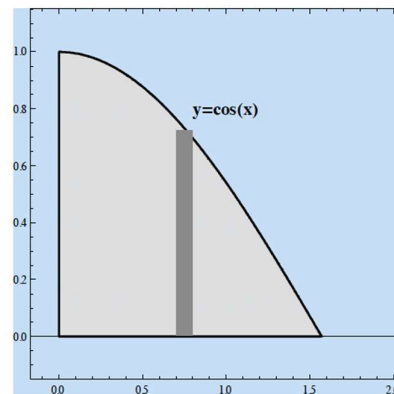
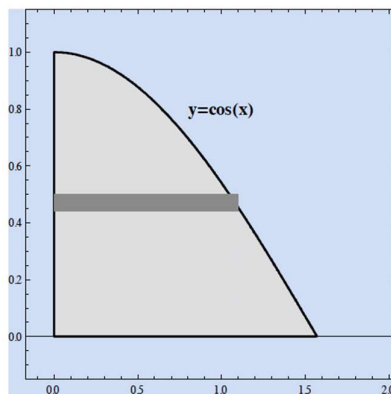
State the equations defining x , y and z in terms of r , θ and ϕ for spherical polar coordinates. Hence evaluate the integral

$$I = \iiint_V z^6 dx dy dz$$

over the volume V contained within the sphere of radius a centred at the origin.

Solution

(a) If $\int_0^1 \int_0^{\arccos(y)} \tan(x) dx dy$ is the integral of the function $\tan(x)$ over the region defined by the lines $y = 0$ and $y = 1$, and $x = 0$ and $x = \arccos(y)$, it is expressed so that the x integration will be carried out first (horizontal segments - left-hand diagram).



Reversing the order of integration (to give vertical strips - right-hand diagram) gives new lower and upper limits for y of the form $y = 0$ and $y = \cos(x)$, and constant limits for x of the form $x = 0$ and $x = \frac{\pi}{2}$.

The integral then becomes

$$I = \int_0^{\pi/2} \int_0^{\cos(x)} \tan(x) dy dx$$

But $\tan(x)$ is a constant as far as the y integration is concerned, so that

$$\begin{aligned}
I &= \int_0^{\frac{\pi}{2}} \int_0^{\cos(x)} \tan(x) \, dy dx = \int_0^{\frac{\pi}{2}} \tan(x) \int_0^{\cos(x)} 1 \, dy dx \\
&= \int_0^{\frac{\pi}{2}} \tan(x) \left(\int_0^{\cos(x)} 1 \, dy \right) dx \\
&= \int_0^{\frac{\pi}{2}} \tan(x) [y]_0^{\cos(x)} dx \\
&= \int_0^{\frac{\pi}{2}} \tan(x) \cos(x) \, dx = \int_0^{\frac{\pi}{2}} \sin(x) \, dx \\
&= [-\cos(x)]_0^{\frac{\pi}{2}} \\
&= 0 + 1 = 1
\end{aligned}$$

(b) The equations defining x , y and z in spherical polar coordinates are:

$$\begin{aligned}
x &= r \sin(\theta) \cos(\phi) \\
y &= r \sin(\theta) \sin(\phi) \\
z &= r \cos(\theta)
\end{aligned}$$

If $I = \iiint_V z^6 dx dy dz$ in Cartesian coordinates, then switching to spherical polar coordinates gives a volume element $dV = r^2 \sin(\theta) dr d\theta d\phi$ and an integrand of $z^6 = (r \cos(\theta))^6$.

For a sphere of radius a , the limits defining the volume region will be

$$\begin{aligned}
0 &\leq r \leq a \\
0 &\leq \theta \leq \pi \\
0 &\leq \phi \leq 2\pi
\end{aligned}$$

Hence, the change of variables will produce the integral

$$I = \int_0^{2\pi} \int_0^\pi \int_0^a \left(r^6 \cos^6(\theta) \right) r^2 \sin(\theta) dr d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^\pi \int_0^a r^8 \cos^6(\theta) \sin(\theta) dr d\theta d\phi$$

$$= \int_0^{2\pi} d\phi \int_0^\pi \cos^6(\theta) \sin(\theta) d\theta \int_0^a r^8 dr$$

$$= \left[\phi \right]_0^{2\pi} \left[-\frac{\cos^7(\theta)}{7} \right]_0^\pi \left[\frac{r^9}{9} \right]_0^a$$

$$= \left[2\pi - 0 \right] \left[\frac{1}{7} - \left(-\frac{1}{7} \right) \right] \left[\frac{a^9}{9} - 0 \right] = \frac{4\pi a^9}{63}$$

PAST EXAMINATION QUESTIONS - 2020

(2020 Stage 2 Examinations, question 1)

(a) [12 Marks]

If a two-variable function $f(x, y)$ is subjected to a transformation of coordinates defined by

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

Express $\frac{\partial f}{\partial r}$ and $\frac{\partial f}{\partial \theta}$ in terms of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Obtain $\frac{\partial f}{\partial r}$ and $\frac{\partial f}{\partial \theta}$ for the particular case when $f(x, y) = x^2y + xe^y$, expressing your answer in terms of r and θ .

(b) [13 Marks]

By reversing the order of integration, evaluate the integral

$$\int_0^1 \int_0^{\arccos(y)} x^2 \sec(x) \, dx \, dy$$

over the region defined by the lines $y = 0$ and $y = 1$, and $x = 0$ and the curve $x = \arccos(y)$.

Sketch plots illustrating the shape of the region of integration and the orientation of the infinitesimal area elements for both the original and the reversed order of integration.

Solution

(a) In general, $\frac{\partial f}{\partial r}$ and $\frac{\partial f}{\partial \theta}$ can be expressed in terms of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ by

$$\begin{pmatrix} \frac{\partial f}{\partial r} \\ \frac{\partial f}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

In the particular case when $x = r \cos(\theta)$ and $y = r \sin(\theta)$ this becomes,

$$\begin{pmatrix} \frac{\partial f}{\partial r} \\ \frac{\partial f}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -r \sin(\theta) & r \cos(\theta) \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

Further, if $f(x, y) = x^2y + xe^y$, then

$$\frac{\partial f}{\partial x} = 2xy + e^y = 2r^2 \cos(\theta) \sin(\theta) + e^{r \sin(\theta)}$$

and
$$\frac{\partial f}{\partial y} = x^2 + xe^y = r^2 \cos^2(\theta) + r \cos(\theta) e^{r \sin(\theta)}$$

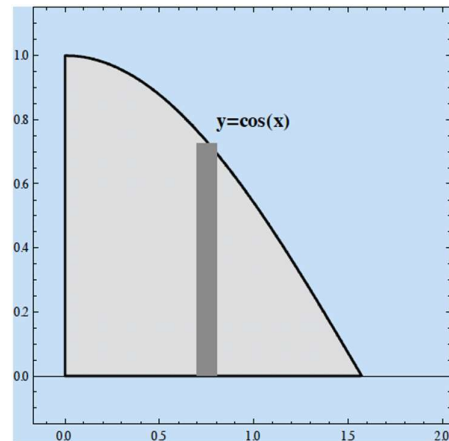
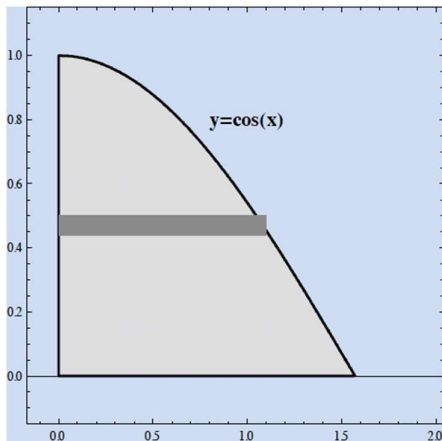
So that

$$\begin{aligned} \frac{\partial f}{\partial r} &= \left(2r^2 \cos^2(\theta) \sin(\theta) + \cos(\theta) e^{r \sin(\theta)} \right) + \left(r^2 \cos^2(\theta) \sin(\theta) + r \cos(\theta) \sin(\theta) e^{r \sin(\theta)} \right) \\ &= 3r^2 \cos^2(\theta) \sin(\theta) + \cos(\theta) (1 + r \sin(\theta)) e^{r \sin(\theta)} \end{aligned}$$

and
$$\begin{aligned} \frac{\partial f}{\partial \theta} &= \left(-2r^3 \cos(\theta) \sin^2(\theta) - r \sin(\theta) e^{r \sin(\theta)} \right) + \left(r^3 \cos^3(\theta) + r^2 \cos^2(\theta) e^{r \sin(\theta)} \right) \\ &= r^3 \cos(\theta) (\cos^2(\theta) - 2 \sin^2(\theta)) + r (r \cos^2(\theta) - \sin(\theta)) e^{r \sin(\theta)} \end{aligned}$$

(b)

If $I = \int_0^1 \int_0^{\arccos(y)} x^2 \sec(x) \, dx dy$ is the integral of the function $x^2 \sec(x)$ over the region defined by the lines $y = 0$ and $y = 1$, and $x = 0$ and $x = \arccos(y)$, it is expressed so that the x integration will be carried out first (horizontal segments – left hand diagram).



Reversing the order of integration (gives vertical strips – right-hand diagram)

This gives new lower and upper limits for y of the form $y = 0$ and $y = \cos(x)$, and constant limits for x of the form $x = 0$ and $x = \frac{\pi}{2}$. The integral then becomes

$$I = \int_0^{\pi/2} \int_0^{\cos(x)} x^2 \sec(x) \, dy dx$$

But $x^2 \sec(x)$ is a constant as far as the y integration is concerned, so that

$$\begin{aligned}
 I &= \int_0^{\pi/2} \int_0^{\cos(x)} x^2 \sec(x) dy dx = \int_0^{\pi/2} x^2 \sec(x) \int_0^{\cos(x)} 1 dy dx \\
 &= \int_0^{\pi/2} x^2 \sec(x) \left(\int_0^{\cos(x)} 1 dy \right) dx \\
 &= \int_0^{\pi/2} x^2 \sec(x) [y]_0^{\cos(x)} dx \\
 &= \int_0^{\pi/2} x^2 \sec(x) \cos(x) dx = \int_0^{\pi/2} x^2 dx = \left[\frac{x^3}{3} \right]_0^{\pi/2} = \frac{\pi^3}{24}
 \end{aligned}$$

(2020 Stage 2 Examinations, question 2)

(a) [12 Marks]

Establish the positions and nature of the stationary points of the function $z = 8x^4 - 4xy + y^2$.

(b) [13 Marks]

Evaluate the integral

$$\iiint_V z^2 dx dy dz$$

over the volume V contained within the positive octant of a sphere of a radius a centred at the origin.

Solution

(a)

If $z = 8x^4 - 4xy + y^2$, then

$$\frac{\partial z}{\partial x} = 32x^3 - 4y \quad \text{and} \quad \frac{\partial z}{\partial y} = -4x + 2y$$

The stationary points are defined when both these expressions are equal to zero:

$$\left. \begin{aligned} 32x^3 - 4y &= 0 \\ \text{and } -4x + 2y &= 0 \end{aligned} \right\} \Rightarrow 32x^3 - 8x = 0 \Rightarrow 8x(4x^2 - 1) = 0$$

where y has been eliminated. This means that possible stationary points could occur when $x = 0, -\frac{1}{2}, \frac{1}{2}$.

Substituting back into the two derivative equations confirms that the corresponding y values would be $x = 0, -1, 1$ and that the three coordinates $(0,0)$, $(-\frac{1}{2}, 1)$, and $(\frac{1}{2}, 1)$ all define stationary points.

Next, differentiating again shows that

$$\frac{\partial^2 z}{\partial x^2} = 96x^2, \quad \frac{\partial^2 z}{\partial y^2} = 2 \quad \text{and} \quad \frac{\partial^2 z}{\partial x \partial y} = -4, \quad \text{so that the discriminant is}$$

$$\Delta = \frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 = 192x^2 - 16.$$

In general, at a stationary point,

if $\Delta > 0$ and $\frac{\partial^2 z}{\partial x^2} < 0$, the point represents a local maximum,

if $\Delta > 0$ and $\frac{\partial^2 z}{\partial x^2} > 0$, the point represents a local minimum,

if $\Delta < 0$ the point represents a saddle point.

In this case, at the point $(0, 0)$, $\Delta = -16 < 0$, so that it is a saddle point.

At the point $\left(-\frac{1}{2}, -1\right)$, $\Delta = 32 > 0$ and $\frac{\partial^2 f}{\partial x^2} = 24 > 0$, so that it is a local minimum.

At the point $\left(\frac{1}{2}, 1\right)$, $\Delta = 32 > 0$ and $\frac{\partial^2 f}{\partial x^2} = 24 > 0$, so that it is a local minimum.

(b)

To evaluate the integral $\iiint_V z^2 dx dy dz$ over a sphere of a radius a centred at the origin, transform to spherical polar coordinates using

$$x = r \sin(\theta) \cos(\phi)$$

$$y = r \sin(\theta) \sin(\phi)$$

$$z = r \cos(\theta)$$

For this transformation:

the volume element is: $dx dy dz = r^2 \sin(\theta) dr d\theta d\phi$

the integrand is: $f(r, \theta, \phi) = r^2 \cos^2(\theta)$

and the limits are defined by the ranges $r \in [0, a]$, $\theta \in \left[0, \frac{\pi}{2}\right]$, and $\phi \in \left[0, \frac{\pi}{2}\right]$

since the integral is over only the one-eighth of the sphere within the positive octant.

Hence,

$$\begin{aligned}
 I &= \iiint_V z^2 dz dy dx = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a \left(r^2 \cos^2(\theta) \right) r^2 \sin(\theta) dr d\theta d\phi \\
 &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^a r^4 \sin(\theta) \cos^2(\theta) dr d\theta d\phi
 \end{aligned}$$

The integrand is a simple product of separate functions with constant limits – the integral can be broken down into three independent parts, giving

$$\begin{aligned}
 I &= \left(\int_0^{\frac{\pi}{2}} d\phi \right) \left(\int_0^{\frac{\pi}{2}} \sin(\theta) \cos^2(\theta) d\theta \right) \left(\int_0^a r^4 dr \right) \\
 &= \left[\phi \right]_0^{\frac{\pi}{2}} \left[\left(-\frac{1}{3} \cos^3(\theta) \right) \right]_0^{\frac{\pi}{2}} \left[\frac{1}{5} r^5 \right]_0^a \\
 &= \left(\frac{\pi}{2} \right) \left(\frac{1}{3} \right) \left(\frac{a^5}{5} \right) = \frac{\pi a^5}{30}
 \end{aligned}$$

PAST EXAMINATION QUESTIONS - 2021

(2021 Stage 2 Examinations, question 1)

(a) [8 Marks]

Let $z = xe^y + \sin(x^2y)$ and $\cos(y^2) = e^x + \ln|x|$ be. State the general expression for the total derivative $\frac{dz}{dy}$, in terms of $\frac{dx}{dy}$ and partial derivatives, $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$. Using this expression, evaluate $\frac{dz}{dy}$ for the given example.

(b) [12 Marks]

Reverse the order of integration, in order to evaluate the integral

$$\iint_A y \sin^2(x) \, dx dy$$

over the area defined by $y = 0$ and $y = 2$, and $x = \frac{1}{2}y^2$ and $x = 2$. The answer must include the sketch of plots that illustrate the shape of the region of integration and the orientation of the infinitesimal area elements, for both the original and the reversed order of integration.

Solution

(a)

When $z = xe^y + \sin(x^2y)$ and $\cos(y^2) = e^x + \ln|x|$, we can say that z is implicitly a function of only y .

In general, if $z = f(x, y)$, $x = g(t)$ and $y = w(t)$, then the chain rules state that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Here, for the case when x and y are related to each other, we can write that the total differential is given by the following expression

$$\frac{dz}{dy} = \frac{\partial z}{\partial x} \frac{dx}{dy} + \frac{\partial z}{\partial y} \frac{dy}{dy} = \frac{\partial z}{\partial x} \frac{dx}{dy} + \frac{\partial z}{\partial y}$$

In this case,

$$\begin{aligned} \frac{\partial z}{\partial x} &= e^y + \cos(x^2y) 2xy \\ \frac{\partial z}{\partial y} &= xe^y + x^2 \cos(x^2y) \end{aligned}$$

and implicit differentiation of $\cos(y^2) = e^x + \ln|x|$ gives

$$-\sin(y^2) 2y = \left(e^x + \frac{1}{x}\right) \frac{dx}{dy} \Rightarrow \frac{dx}{dy} = -\frac{2y \sin(y^2)}{e^x + 1/x}$$

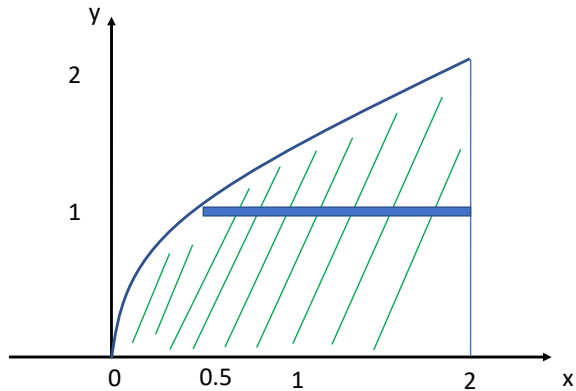
Thus, we have

$$\frac{dz}{dy} = -(e^y + 2xy \cos(x^2 y)) \left(\frac{2y \sin(y^2)}{e^x + \frac{1}{x}} \right) + x e^y + x^2 \cos(x^2 y)$$

(b)

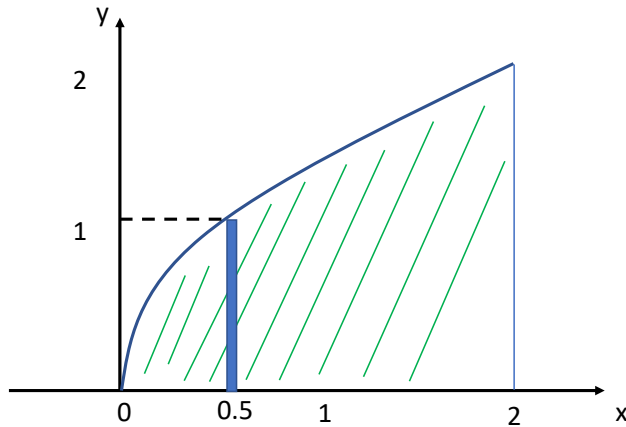
This is a double integral that represents an integration over a region, which is defined by the curve $x = \frac{1}{2}y^2$, $x = 4$, and the x -axis.

Note that the integration is carried out first, corresponding to horizontal infinitesimal area elements, for any given y :



The above area is obtained by evaluating $I = \int_0^2 \int_{y^2/2}^2 y \sin^2(x) \, dx dy$.

Reverse the order of integration will produce the integral $I = \int_0^2 \int_0^{\sqrt{2x}} y \sin^2(x) \, dy dx$. Here, the y -integration is carried out first, corresponding to vertical infinitesimal area elements, and the limits are defined by the curve $y = 0$, and $y = \sqrt{2x}$:



Based on these,

$$\begin{aligned}
 I &= \int_0^2 \int_0^{\sqrt{2x}} y \sin^2(x) \, dx dy = \int_0^2 \sin^2(x) \left(\int_0^{\sqrt{2x}} y \, dy \right) dx \\
 &= \int_0^2 \sin^2(x) \left[\frac{1}{2} y^2 \right]_0^{\sqrt{2x}} dx = \int_0^2 x \sin^2(x) \, dx \\
 &= \int_0^2 x \left(\frac{1}{2} - \cos(2x) \right) dx = 1 - \frac{1}{2} \int_0^2 x \cos(2x) \, dx
 \end{aligned}$$

Here, let $u = 2x$ and $du = 2x dx$. Using these, the second term on the right side in the last equality can be represented by

$$\int_0^4 \frac{u}{2} \cos(u) \, du \cdot \frac{1}{2} = \frac{1}{4} \{4 \sin(4) + \cos(4) - 1\}$$

Accordingly, we will have

$$I = \frac{9}{8} - \frac{1}{8} \{4 \sin(4) + \cos(4)\} = 1.5851$$

(2021 Stage 2 Examinations, question 2)

(a) [10 Marks]

Consider a two-dimensional surface, represented by $z = 2x^2 - 2xy + x^3 + xy^2$. Obtain the coordinates and nature of the stationary points of the surface.

(b) [10 Marks]

Denote by $f(x, y)$ a two-variable function. State the general expression for a Taylor series expansion of $f(x, y)$ as far as terms of order two, and use this generalised expression in order to obtain a second-order approximation at the point $\left(0, \frac{\pi}{4}\right)$ for the function

$$f(x, y) = e^{-xy} \sin(y).$$

Solution

(a)

If $z = 2x^2 - 2xy + x^3 + xy^2$, then

$$f_x = \frac{\partial z}{\partial x} = 4x - 2y + 3x^2 + y^2 \text{ and } f_x = 0. \Rightarrow y^2 - 2y + 3x^2 + 4x = 0 \text{ (condition 1)}$$

$$\text{And } f_y = \frac{\partial z}{\partial y} = -2x + 2xy \text{ and } f_y = 0. \Rightarrow 2x(y - 1) = 0. \Rightarrow x = 0 \text{ or } y = 1 \\ \text{(condition 2)}$$

(i) Considering $x = 0$ (from condition 2) into condition 1, we obtain:

$$f_x = y^2 - 2y = 0 \Rightarrow y = 0 \text{ or } y = 2$$

(ii) Considering $y = 1$ (from condition 2) into condition 1, we have:

$$f_y = 3x^2 + 4x - 1 = 0 \Rightarrow \left(x + \frac{2}{3}\right)^2 = \frac{7}{9} \Rightarrow x = \frac{-2-\sqrt{7}}{3} \text{ or } x = \frac{-2+\sqrt{7}}{3}$$

Based on these, there are four stationary points (x, y) at:

$$(0, 0), (0, 2), \left(\frac{-2-\sqrt{7}}{3}, 1\right), \left(\frac{-2+\sqrt{7}}{3}, 1\right)$$

As for the nature of these stationary points:

$$\text{At } (0, 0), \text{ the value of the discriminants is } \Delta = \begin{vmatrix} 4 & -2 \\ -2 & 0 \end{vmatrix} < 0.$$

Since $\Delta < 0$, the stationary point at $(0, 0)$ is a saddle point.

$$\text{At } (0, 2), \text{ the value of the discriminants is } \Delta = \begin{vmatrix} 4 & 2 \\ 2 & 4 \end{vmatrix} = 16 - 4 = 12 > 0.$$

Since $\Delta > 0$ and $f_{xx} = \frac{\partial^2 f}{\partial x^2} > 0$, the stationary point at $(0, 2)$ is a local minimum.

$$\text{At } \left(\frac{-2-\sqrt{7}}{3}, 1\right), \text{ the value of the discriminants is } \Delta = \begin{vmatrix} -5.2915 & 0 \\ 0 & -0.43050 \end{vmatrix} > 0.$$

Since $\Delta > 0$ and $f_{xx} = \frac{\partial^2 f}{\partial x^2} < 0$, the stationary point at $\left(\frac{-2-\sqrt{7}}{3}, 1\right)$ is a local maximum.

$$\text{At } \left(\frac{-2+\sqrt{7}}{3}, 1\right), \text{ the value of the discriminants is } \Delta = \begin{vmatrix} 5.2915 & 0 \\ 0 & 0.43050 \end{vmatrix} > 0.$$

Since $\Delta > 0$ and $f_{xx} = \frac{\partial^2 f}{\partial x^2} > 0$, the stationary point at $\left(\frac{-2+\sqrt{7}}{3}, 1\right)$ is a local minimum.

(b)

At any given point (a, b) , the value of a two-variable function can be approximated by using a Taylor series expansion for the two variable function. The general expressions of the Taylor series expansion is:

$$f(x, y) = f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b) + \frac{1}{2!}(x - a)^2 f_{xx}(a, b) + \frac{1}{2!}(y - b)^2 f_{yy}(a, b) + (x - a)(y - b)f_{xy}(a, b) + \dots$$

If $f(x, y) = e^{-xy} \sin(y)$, then $f\left(0, \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$f_x = \sin(y) e^{-xy}(-y)$, so that $f_x\left(0, \frac{\pi}{4}\right) = -\frac{\pi}{4\sqrt{2}}$

$f_y = (-x)e^{-xy} \sin(y) + e^{-xy} \cos(y)$, so that $f_y\left(0, \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$f_{xx} = y^2 \sin(y) e^{-xy}$, so that $f_{xx}\left(0, \frac{\pi}{4}\right) = \frac{\pi^2}{4\sqrt{2}}$

$f_{yy} = e^{-xy} x^2 \sin(y) - 2xe^{-x} \cos(y) - e^{-x} \sin(y)$, so that $f_{yy}\left(0, \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

$f_{xy} = -ye^{-xy} \cos(y) + \sin(y) e^{-xy}(xy - 1)$, so that $f_{xy}\left(0, \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}\left(1 + \frac{\pi}{4}\right)$

Therefore, we have

$$f(x, y) \approx \frac{1}{\sqrt{2}} - \frac{\pi}{4\sqrt{2}}x + \frac{1}{\sqrt{2}}\left(y - \frac{\pi}{4}\right) + \frac{\pi^2}{8\sqrt{2}}x^2 - \frac{1}{2\sqrt{2}}\left(y - \frac{\pi}{4}\right)^2 - x\left(y - \frac{\pi}{4}\right)\left(\frac{1 + \frac{\pi}{4}}{\sqrt{2}}\right)$$

PAST EXAMINATION QUESTIONS - 2022

(2022 Stage 2 Examinations, question 1)

(a)

If two coordinate systems are related by the transformation:

$$x(u, w), \quad y(u, w) \rightarrow u(x, y), \quad w(x, y)$$

then the equations that convert gradient information from the xy coordinate system to the uw coordinate system are

$$\begin{aligned}\frac{\partial f}{\partial u} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ \frac{\partial f}{\partial w} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial w} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial w}\end{aligned}$$

The question is asking for the first of these equations to be evaluated. Note that we cannot directly work out $\frac{\partial x}{\partial u}$ and $\frac{\partial y}{\partial u}$ since we are given the transformation from (u, w) to (x, y) , not the other way.

$$\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial w}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial u} \\ \frac{\partial f}{\partial w} \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\partial f}{\partial u} \\ \frac{\partial f}{\partial w} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial w}{\partial y} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

In this case,

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial w}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \exp(x^2 + y^2) 2x & 1/y \\ \frac{1}{2} \exp(x^2 + y^2) 2y & -x/y^2 \end{pmatrix}$$

The determinant of this matrix is

$$\left(\frac{1}{2} \exp(x^2 + y^2) 2x \right) \left(-\frac{x}{y^2} \right) - \left(\frac{1}{2} \exp(x^2 + y^2) 2y \right) \left(\frac{1}{y} \right) = -\exp(x^2 + y^2) \left(\frac{x^2}{y^2} + 1 \right)$$

So, the inverse is given by

$$\begin{aligned} \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial w}{\partial y} \end{pmatrix}^{-1} &= \frac{-y^2}{(x^2 + y^2) \exp(x^2 + y^2)} \begin{pmatrix} -x/y^2 & -1/y \\ -\frac{1}{2} \exp(x^2 + y^2) 2y & \frac{1}{2} \exp(x^2 + y^2) 2x \end{pmatrix} \\ &= \begin{pmatrix} \frac{x}{(x^2 + y^2) \exp(x^2 + y^2)} & \frac{y}{(x^2 + y^2) \exp(x^2 + y^2)} \\ \frac{y^3}{x^2 + y^2} & -\frac{xy^2}{x^2 + y^2} \end{pmatrix} \end{aligned}$$

Therefore, we have

$$\begin{pmatrix} \frac{\partial f}{\partial u} \\ \frac{\partial f}{\partial w} \end{pmatrix} = \begin{pmatrix} \frac{x}{(x^2 + y^2) 2u} & \frac{y}{(x^2 + y^2) 2u} \\ \frac{y^3}{x^2 + y^2} & -\frac{xy^2}{x^2 + y^2} \end{pmatrix} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

and multiplying out just the first line of this matrix-vector equation provides

$$\frac{\partial f}{\partial u} = \frac{1}{2u(x^2 + y^2)} \left(\frac{\partial f}{\partial x} x + \frac{\partial f}{\partial y} y \right)$$

as requested.

(b)

Given a two-variable function f , the total change in f due to small changes of both input variables x and y can be approximated using the partial derivatives in the directions of increasing x and y as:

$$\Delta f \approx \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

where Δx (or Δy) denotes the small change in x (or y).

In this case, there exist four possible points for (x_2, y_2) by changing $\Delta x \in \{+\alpha, -\alpha\}, \Delta y \in \{+\alpha, -\alpha\}$. Here, Δf indicates the total change in f due to the small changes of input variables by $\pm\alpha$.

Note that

$$\begin{aligned} \frac{\partial f}{\partial x} = f_x &= \frac{\partial}{\partial x} (\exp(-x^2 y) + x) = -2xy \exp(-x^2 y) + 1 \\ \frac{\partial f}{\partial y} = f_y &= \frac{\partial}{\partial y} (\exp(-x^2 y) + x) = -x^2 \exp(-x^2 y) \end{aligned}$$

In particular, at the point $(1, 4)$, we have

$$\begin{aligned} f_x @ (1, 4) &= -8 \exp(-4) + 1 > 0 \\ f_y @ (1, 4) &= -\exp(-4) < 0 \end{aligned}$$

Given these, to obtain a steepest increase of f , we must change the input variables to a point which is "in the direction of increasing at most the total change of f ." In this light, when (x_1, y_1) is allowed to change to (x_2, y_2) as one of 4 possible coordinate points: $(x_2, y_2) = (1 \pm \alpha, 4 \pm \alpha)$, the steepest increase of f is obtained by maximizing Δf over the possible next points.

When having $\Delta f @ (1, 4) \rightarrow (1 + \alpha, 4 - \alpha)$, the corresponding total change is

$$\Delta f \approx f_x \times (\alpha) + f_y \times (-\alpha) = 0.8718$$

Therefore, $(x_2, y_2) = (x_1 + \alpha, y_1 - \alpha) = (2, 3)$ leads to the steepest increase of Δf .

(2022 Stage 2 Examinations, question 2)

(a)

To solve the integral over the half sphere of radius $a = 2$ centred at the origin, we firstly transform to spherical polar coordinates using the following:

$$\begin{aligned} x &= r \sin(\theta) \cos(\phi) \\ y &= r \sin(\theta) \sin(\phi) \\ z &= r \cos(\theta) \end{aligned}$$

Note that we have

$$(i) \quad \text{The volume element: } dx dy dz = r^2 \sin(\theta) dr d\theta d\phi$$

- (ii) The integrand: $f(r, \theta, \phi) = r^2 \cos^2(\theta)$
 (iii) The limits defined by the ranges: $r \in [0, a]$, $\theta \in [0, \pi]$, and $\phi \in [0, \pi]$

The integral can be rewritten in the spherical polar coordinates:

$$\begin{aligned} I &= \iiint_V z^2 \, dx dy dz = \int_0^\pi \int_0^\pi \int_0^{a=2} (r^2 \cos^2(\theta)) r^2 \sin(\theta) \, dr d\theta d\phi \\ &= \int_0^\pi \int_0^\pi \int_0^{a=2} r^4 \cos^2(\theta) \sin(\theta) \, dr d\theta d\phi \\ &= \int_0^\pi \int_0^\pi \sin(\theta) \cos^2(\theta) \int_0^{a=2} r^4 \, dr d\theta d\phi \end{aligned}$$

The integrand is a simple product of separate functions with constant limits. Accordingly, the integral can be broken down into three independent integrals, producing

$$\begin{aligned} I &= \left\{ \int_0^\pi d\phi \right\} \left\{ \int_0^\pi \sin(\theta) \cos^2(\theta) \, d\theta \right\} \left\{ \int_0^{a=2} r^4 \, dr \right\} \\ &= [\phi]_0^\pi \left[-\frac{1}{3} \cos^3(\theta) \right]_0^\pi \left[\frac{1}{5} r^5 \right]_0^{a=2} \\ &= (\pi) \left(\frac{1}{3} + \frac{1}{3} \right) \left(\frac{32}{5} \right) = \frac{\pi 64}{15} \end{aligned}$$

(b)

Consider the double integral in the Cartesian coordinates:

$$I = \int_0^{b/2} \int_{2x}^{\sqrt{b^2-x^2}} (x^2 + y^2)^{\frac{1}{2}} \, dy dx$$

To transform it to polar coordinates, we use

$$\begin{aligned} x &= r \cos(\theta) \\ y &= r \sin(\theta) \end{aligned}$$

The Jacobian of the transformation is obtained as

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{vmatrix} = r \cos^2(\theta) + r \sin^2(\theta) = r$$

Using the Jacobian can provide that $dx dy = r \, dr d\theta$.

This can be quoted as a standard result for 2D polar coordinates.

The integration in terms of y contains a lower limit by the line $y = 2x$ and an upper limit by the curve $y = \sqrt{b^2 - x^2}$. The upper limit is related to $x^2 + y^2 = b^2$ and thus, the interval for x can be given as $x \in \left[0, \frac{b}{2}\right]$

The integration region is one-sixteenth of a circle of radius b . Thus, the new limits of integration in the polar coordinate system are

- The radial variable: $r \in [0, b]$
- The angular variable: $\theta \in \left[\frac{3\pi}{8}, \frac{\pi}{2}\right]$

Based on this transformation, we have

$$I = \int_0^{\frac{b}{2}} \int_{2x}^{\sqrt{ba^2-x^2}} (x^2 + y^2)^{\frac{1}{2}} dy dx = \int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \int_0^b (r^1) r dr d\theta$$

Performing the inner and external integrals gives

$$\begin{aligned} I &= \int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \int_0^b r^2 dr d\theta = \int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \left[\frac{1}{3} r^3 \right]_0^b d\theta \\ &= \int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \frac{b^3}{3} d\theta = \frac{b^3}{3} \int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} 1 d\theta = \frac{b^3}{3} [\theta]_{\frac{3\pi}{8}}^{\frac{\pi}{2}} = \frac{b^3}{3} \left(\frac{\pi}{2} - \frac{3\pi}{8} \right) = \frac{\pi}{24} b^3 \end{aligned}$$

PAST EXAMINATION QUESTIONS - 2023

(2023 Stage 2 Examinations, question 1)

(a)

We consider a transformation to a spherical Polar Coordinates using the conversion formula between the 3D Cartesian and the 3D Polar Coordinate systems, as

$$\begin{aligned}x &= r \sin(\theta) \cos(\phi) \\y &= r \sin(\theta) \sin(\phi) \\z &= r \cos(\theta)\end{aligned}$$

So, using the Jacobian and the above formula, we can relate the infinitesimal volume elements between the coordinate systems as

$$dx dy dz = r^2 \sin(\theta) dr d\theta d\phi$$

Using these, $f(x, y, z)$ can be reformulated in terms of r, θ , and ϕ as

$$f(r, \theta, \phi) = r^2 \sin^2(\theta) \cos^2(\phi) + r^2 \sin^2(\theta) \sin^2(\phi) = r^2 \sin^2(\theta)$$

with new limits defined by

$$r \in [0, 1], \theta \in \left[0, \frac{\pi}{2}\right], \phi \in \left[0, \frac{\pi}{2}\right]$$

Hence, the 3-dimensional integral can be transformed to the 3D spherical polar coordinate system as:

$$\begin{aligned}I &= \iiint_V x^2 dz dy dx = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^1 r^2 \sin^2(\theta) \cos^2(\phi) r^2 \sin(\theta) dr d\theta d\phi \\&= \int_0^{\frac{\pi}{2}} 1 d\phi \int_0^{\frac{\pi}{2}} \sin^3(\theta) d\theta \int_0^1 r^4 dr \\&= \left[\int_0^{\frac{\pi}{2}} 1 d\phi \right] \left[\int_0^{\frac{\pi}{2}} \sin(\theta) \sin^2(\theta) d\theta \right] \left[\frac{1}{5} \{r^5\}_0^1 \right] \\&= \left[\{\phi\}_0^{\frac{\pi}{2}} \right] \left[\int_0^{\frac{\pi}{2}} \sin(\theta) - \sin(\theta) \cos^2(\theta) d\theta \right] \frac{1}{5}\end{aligned}$$

The integral in terms of θ in the middle can be solved by using $u = \cos(\theta)$:

$$\begin{aligned}u &= \cos(\theta) \rightarrow du = -\sin(\theta) d\theta \\ \theta: 0 \sim \pi/2 &\rightarrow u: 1 \sim 0\end{aligned}$$

Then, the integral for θ can be rewritten as

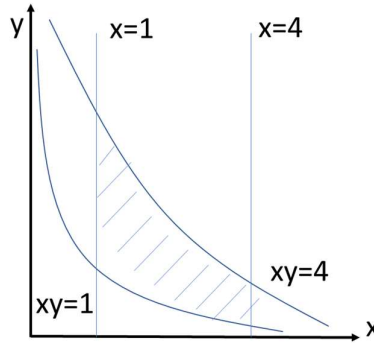
$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin(\theta) d\theta - \int_0^{\frac{\pi}{2}} \sin(\theta) \cos^2(\theta) d\theta &= \int_0^{\frac{\pi}{2}} \sin(\theta) d\theta - \int_0^1 u^2 du \\&= \int_0^{\frac{\pi}{2}} \sin(\theta) d\theta - \frac{1}{3} \{u^3\}_0^1 \\&= \{-\cos(\theta)\}_0^{\pi/2} - \frac{1}{3} = 1 - \frac{1}{3} = \frac{2}{3}\end{aligned}$$

Based on these, we can have

$$\begin{aligned}
 I &= \frac{\pi}{2} \times \left[-\cos(\theta) \right]_0^{\frac{\pi}{2}} + \left[\frac{1}{3} u^3 \right]_0^1 \times \frac{1}{5} \\
 &= \frac{\pi}{2} \times \left(1 - \frac{1}{3} \right) \times \frac{1}{5} = \frac{\pi}{15}
 \end{aligned}$$

(b)

The integral has a variable limit in the integration of y , for given x . This may produce a difficult computation of the integration.



Thus, we may transform the double integral from the (x,y) coordinate system to a new (u,v) coordinate system determining, e.g., the following conversion formula:

$$\begin{aligned}
 x &= \frac{u}{v} \\
 y &= uv
 \end{aligned}$$

To involve all the hyperbola forms between $xy = 1$ and $xy = 4$, we can have $xy = u^2$ such that we transform all the two curves between $xy = 1$ and $xy = 4$:

$$xy = 1 \rightarrow u^2 = 1 \rightarrow u = 1$$

$$xy = 4 \rightarrow u^2 = 4 \rightarrow u = 2$$

into all the lines between $u = 1$ and $u = 2$ in the new coordinate system, where $u \in [1, 2]$

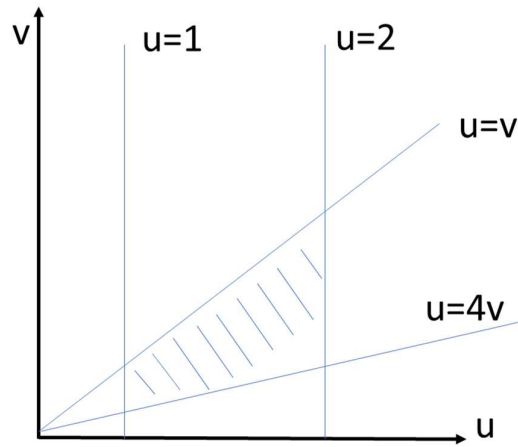
To involve all the lines between $x = 1$ and $x = 4$, we can consider $x = u/v$ s.t.

$$x = 1 \rightarrow \frac{u}{v} = 1 \rightarrow v = u$$

$$x = 4 \rightarrow \frac{u}{v} = 4 \rightarrow v = \frac{1}{4}u$$

where the two straight lines in terms of u and v can express the lines between $x = 1$ and $x = 4$.

These can be used to provide an integration region in the (u,v) coordinates, which is illustrated as



Using the Jacobian and the above formula, we relate the infinitesimal area elements between the coordinate systems as

$$dxdy = \frac{2u}{v} dv du$$

Accordingly, the integral can be rewritten as

$$I = \int_1^2 \int_{\frac{u}{4}}^u (v+u) \frac{2u}{v} dv du$$

where $\sqrt{\frac{y}{x}} = \sqrt{v^2} = v$ and $\sqrt{xy} = \sqrt{u^2} = u$

$$\begin{aligned} I &= \int_1^2 2 \left(\int_{u/4}^u \left(u + u^2 \frac{1}{v} \right) dv \right) du \\ &= \int_1^2 2 \left(u \left[v \right]_{\frac{u}{4}}^u + u^2 \left[\ln|u| - \ln \left| \frac{u}{4} \right| \right] \right) du \\ &= 2 \int_1^2 u^2 \left(\frac{3}{4} + \ln 4 \right) du \\ &= \left(\frac{3}{2} + 2 \ln 4 \right) \int_1^2 u^2 du \\ &= \left(\frac{3}{2} + 2 \ln 4 \right) \left[\frac{1}{3} u^3 \right]_1^2 = \frac{7}{2} + \frac{14}{3} \ln 4 \end{aligned}$$

(2023 Stage 2 Examinations, question 2)

(a)

Note that local maximum points are considered as stationary points. It is therefore needed to find the stationary points of the given two-variable function using its partial derivatives.

In particular, for a given z , we can have

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (3x^2 - 12xy + y^3 + 16) = 6x - 12y$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}(3x^2 - 12xy + y^3 + 16) = -12x + 3y^2$$

Equating the partial derivatives to zero, we identify the conditions for all the stationary points to satisfy, and which are given by:

$$\frac{\partial z}{\partial x} = 0 \rightarrow x - 2y = 0. \quad (\text{Condition 1})$$

$$\frac{\partial z}{\partial y} = 0 \rightarrow 4x - y^2 = 0. \quad (\text{Condition 2})$$

If $x = 2y$, then we can have $y^2 - 8y = 0 \rightarrow y(y - 8) = 0 \rightarrow y = 0$ or $y = 8$.
Accordingly, we can find two possible stationary points at $(0, 0)$ and $(16, 8)$

To find the local maximum point(s), we determine the nature of these points by computing the discriminant as

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= 6 \\ \frac{\partial^2 z}{\partial y^2} &= 6y \\ \frac{\partial^2 z}{\partial xy} &= \frac{\partial^2 z}{\partial yx} = -12 \end{aligned}$$

$$\Delta = \begin{vmatrix} 6 & -12 \\ -12 & 6y \end{vmatrix} = 36y - 144$$

Thus, we can have

@ $\Delta(0,0) = -144 < 0$: Saddle point

@ $\Delta(16,8) = 36 \times 8 - 144 = 144 > 0$ and $\frac{\partial^2 z}{\partial x^2} > 0$: Local minimum point.

Based on these, we can conclude that there is no local maximum point and all the stationary points are saddle point and local minimum point.

(b)

The general expression for a two-variable Taylor series expansion for a reference point (a, b) is:

$$\begin{aligned} f(x, y) &= f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b) \\ &\quad + \frac{1}{2!}(x - a)^2 f_{xx}(a, b) + (x - a)(y - b)f_{xy}(a, b) + \frac{1}{2!}(y - b)^2 f_{yy}(a, b) \\ &\quad + \dots \end{aligned}$$

Using the D operator, alternatively, the general expression can be given by

$$f(a + h, b + k) = f(a, b) + Df(a, b) + \frac{1}{2!}D^2f(a, b) + \frac{1}{3!}D^3f(a, b) + \dots$$

where $D = h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}$ denotes the D operator.

Considering $(a, b) = (\frac{\pi}{4}, 0)$, for a given function, we have

$$f(a, b) = \sin\left(\frac{\pi}{4}\right) = 1/\sqrt{2}$$

$$Df = \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) \sin(x + y) = h \cos(x + y) + k \cos(x + y)$$

$$\begin{aligned} D^2f &= \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) (h + k) \cos(x + y) \\ &= (h + k)h(-\sin(x + y)) + (h + k)k(-\sin(x + y)) \\ &= -(h + k)^2 \sin(x + y) \end{aligned}$$

Using these and the general expression, then, we can have

$$f(a + h, b + k) \approx \sin\left(\frac{\pi}{4}\right) + (h + k) \cos\left(\frac{\pi}{4} + 0\right) + \frac{-1}{2!} (h + k)^2 \sin\left(\frac{\pi}{4} + 0\right)$$

Note that for given $(a, b) = \left(\frac{\pi}{4}, 0\right)$, we can determine

$$\begin{aligned} x &= a + h = \frac{\pi}{4} + h \\ y &= b + k = 0 + k \end{aligned}$$

From these, we further find

$$h + k = x - \frac{\pi}{4} + y$$

and finally, we obtain the Taylor series expansion up to the terms of order two:

$$f(x, y) \approx \frac{1}{\sqrt{2}} + \left(x - \frac{\pi}{4} + y\right) \frac{1}{\sqrt{2}} - \frac{1}{2\sqrt{2}} \left(x - \frac{\pi}{4} + y\right)^2$$
